A Key Transcription Factor Family: Nuclear Receptors

Chapter 5 of **A Key Transcription Factor Family: Nuclear Receptors**, Carsten Carlberg (2024)

April 02 2026

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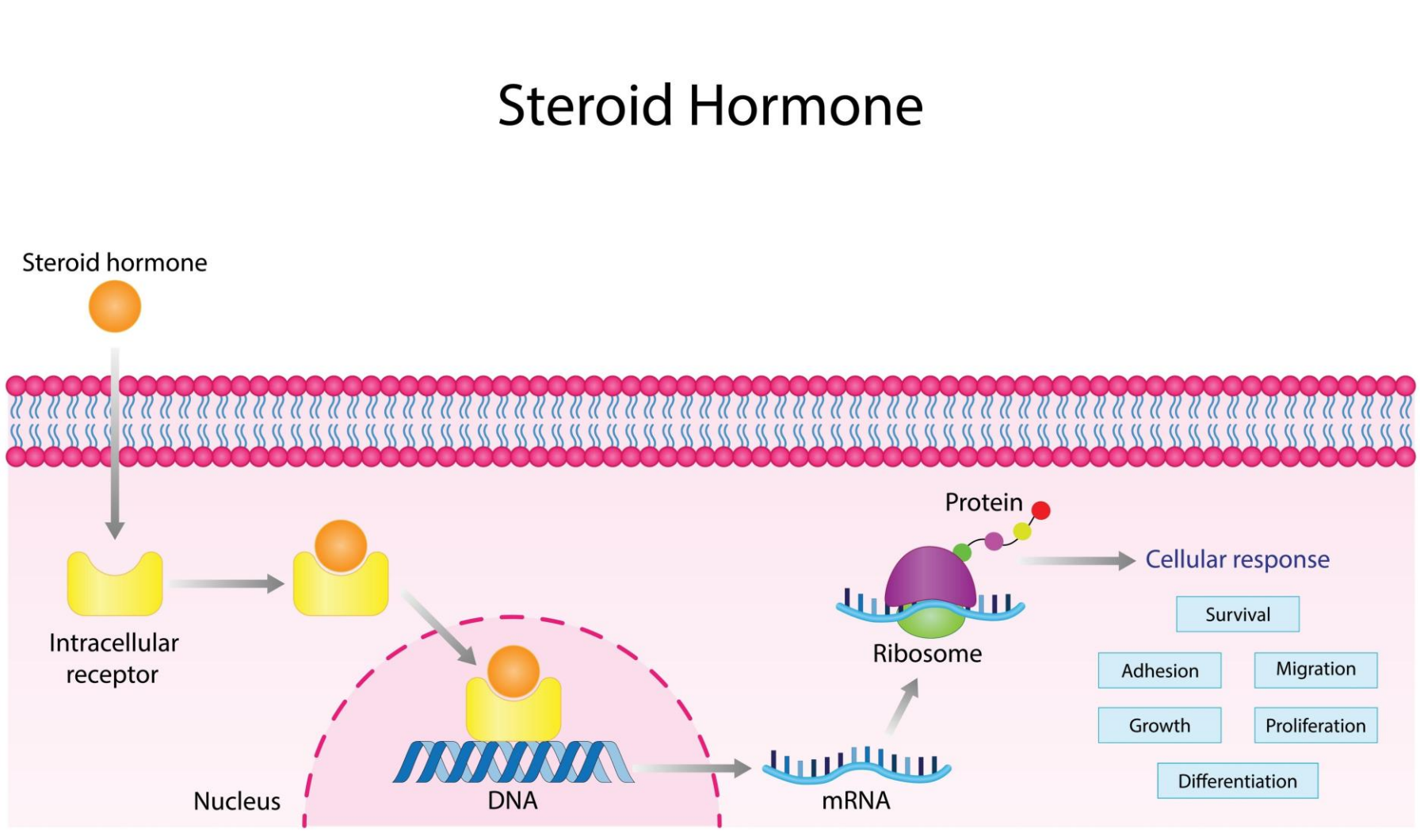
Introduction to Nuclear Receptors (NR)

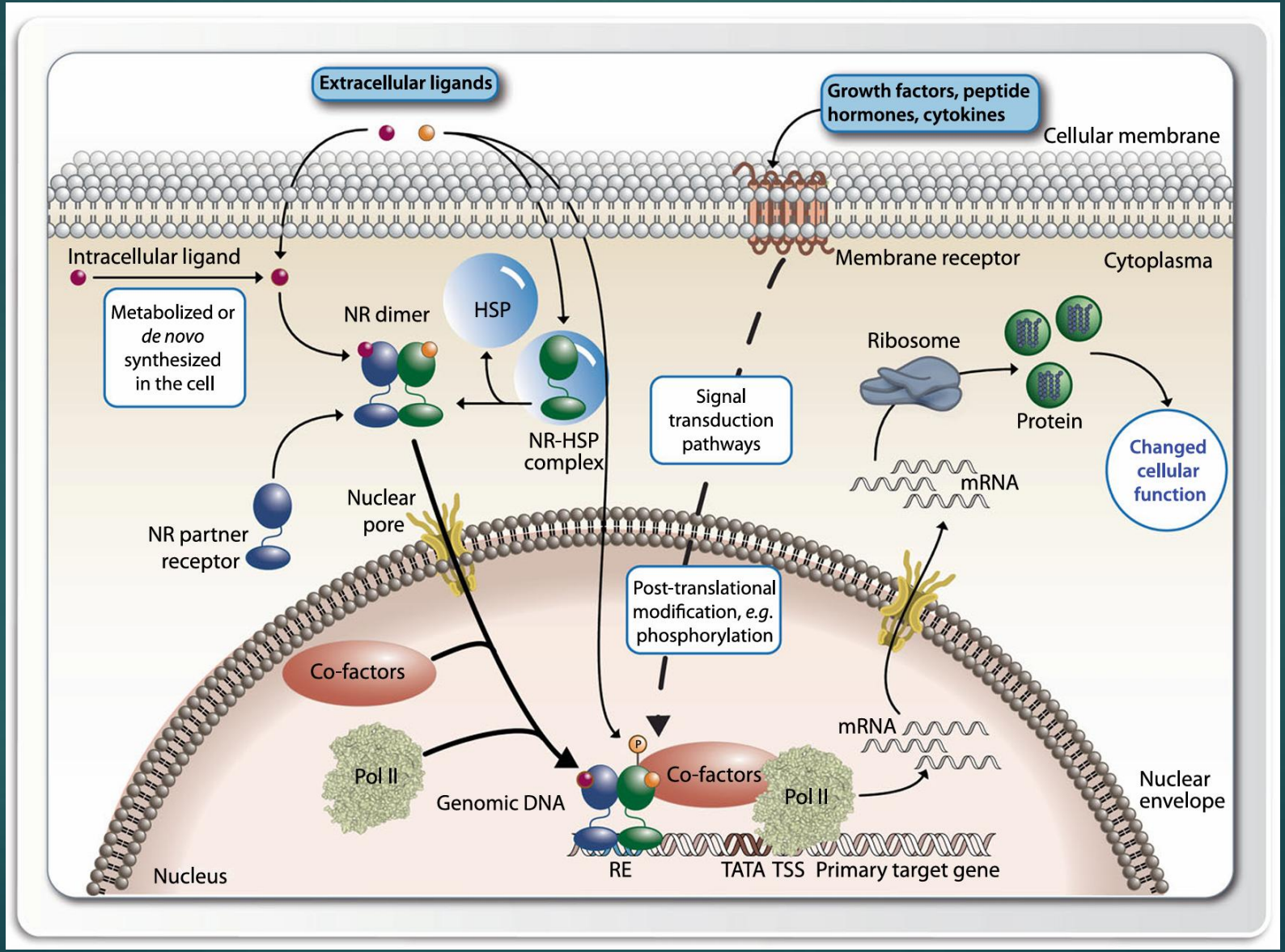


- Overview
- Steroid Hormone Diffuses Into Cell
- Intracellular Receptors
- Conformational Change
- DNA Binding
- MRNA Transcription
- MRNA Translation
- Protein Synthesis
- Cellular Response

Introduction to Nuclear Receptors

- **Definition**: Nuclear receptors are a superfamily of ligand-inducible transcription factors.
- **Significance**: They serve as paradigms for understanding transcription factor function, particularly molecular interactions with coactivators and corepressors.
- **Mechanism**: Unlike hydrophilic signals (e.g., peptide hormones) that require membrane receptors and kinase cascades, nuclear receptors interact with small lipophilic signaling molecules that can pass through the cellular membrane.
- **Direct Link**: They act as direct linkers between extracellular signals and gene transcription, often residing in the nucleus or translocating there upon activation.

Principles of nuclear receptor signaling



The Nuclear Receptor Superfamily: Evolution

- Gene Count: The human genome contains 48 protein-coding genes for nuclear receptors.
- Evolutionary History:
  - Simple eukaryotes (e.g., fungi) lack nuclear receptors.
  - They likely evolved in metazoans as environmental sensors for nutrients and toxins.

The Nuclear Receptor Superfamily: Classification

Human nuclear receptors

Endocrine receptors	AR (NR3C4)	ESR1 (NR3A1)	ESR2 (NR3A2)	GR (NR3C1)
	MR (NR3C2)	PGR (NR3C3)	RARα (NR1B1)	RARβ (NR1B2)
	RARγ (NR1B3)	THRα (NR1A1)	THRβ (NR1A2)	VDR (NR1I1)
Adopted orphan receptors	CAR (NR1I3)	ERRα (NR3B1)	ERRβ (NR3B2)	ERRγ (NR3B3)
	FXR (NR1H4)	HNFα (NR2A1)	HNFγ (NR2A2)	LRH-1 (NR5A2)
	LXRα (NR1H3)	LXRβ (NR1H2)	NGFI-B (NR4A1)	NURR1 (NR4A2)
	PPARα (NR1C1)	PPARδ (NR1C2)	PPARγ (NR1C3)	PXR (NR1I2)
	REV-ERBα (NR1D1)	REV-ERBβ (NR1D2)	RORα (NR1F1)	RORβ (NR1F2)
	RORγ (NR1F3)	RXRα (NR2B1)	RXRβ (NR2B2)	RXRγ (NR2B3)
	SF1 (NR5A1)	TR2 (NR2C1)	TR4 (NR2C2)	
Orphan receptors	COUP-TF1 (NR2F1)	COUP-TF2 (NR2F2)	DAX1 (NR0B1)	EAR2 (NR2F6)
	GCNF (NR6A1)	NOR1 (NR4A3)	PNR (NR2E3)	SHP (NR0B2)
	TLX (NR2E2)			

The Nuclear Receptor Superfamily: Classification

Three Classes

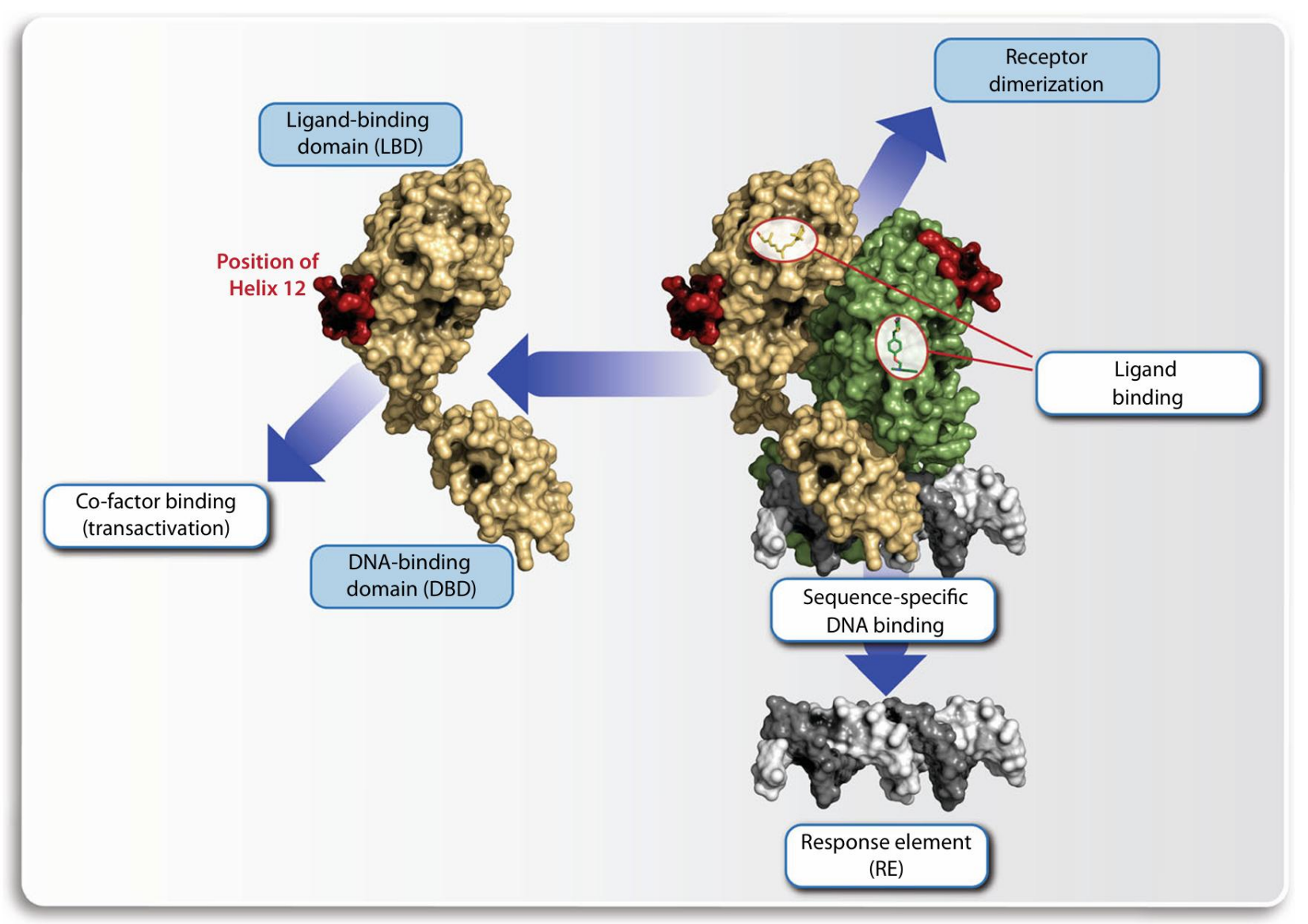
- **Endocrine Receptors (12 members)**: High-affinity receptors (K_d in nanomolar range) for hormones like steroids (estrogen, testosterone, cortisol), Vitamin A/D derivatives, and thyroid hormone (T_3).
- **Adopted Orphan Receptors**: Receptors whose ligands were identified later. Ligands are often dietary lipids/metabolites (fatty acids, bile acids) with lower affinity (K_d in micromolar/millimolar range).
- **True Orphan Receptors**: Receptors with no known natural ligand. They often have small ligand-binding pockets and function like constitutive transcription factors regulated by posttranslational modifications.

Structural Domains of Nuclear Receptor

Key Domains

- **DNA-Binding Domain (DBD):**
  - Highly conserved (66–70 amino acids).
  - Contains two Cys4-type zinc fingers that recognize specific DNA sequences.
- **Ligand-Binding Domain (LBD):**
  - Structurally conserved 3-layer sandwich of 11–13 alpha-helices.
  - Contains the hydrophobic ligand-binding pocket.
- **Helix 12:** A dynamic C-terminal helix that acts like a "mouse trap," flipping position upon ligand binding to stabilize the molecule.
- **N-Terminal Domain:**
  - Variable length (20–450 residues)
  - site for ligand-independent activation via posttranslational modifications.

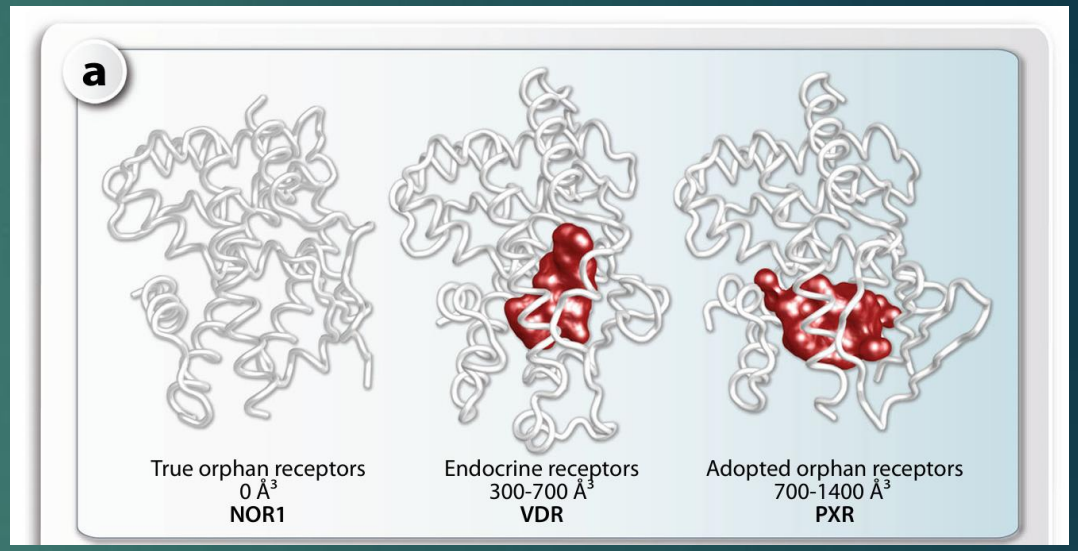
Different molecular interactions of a nuclear receptor



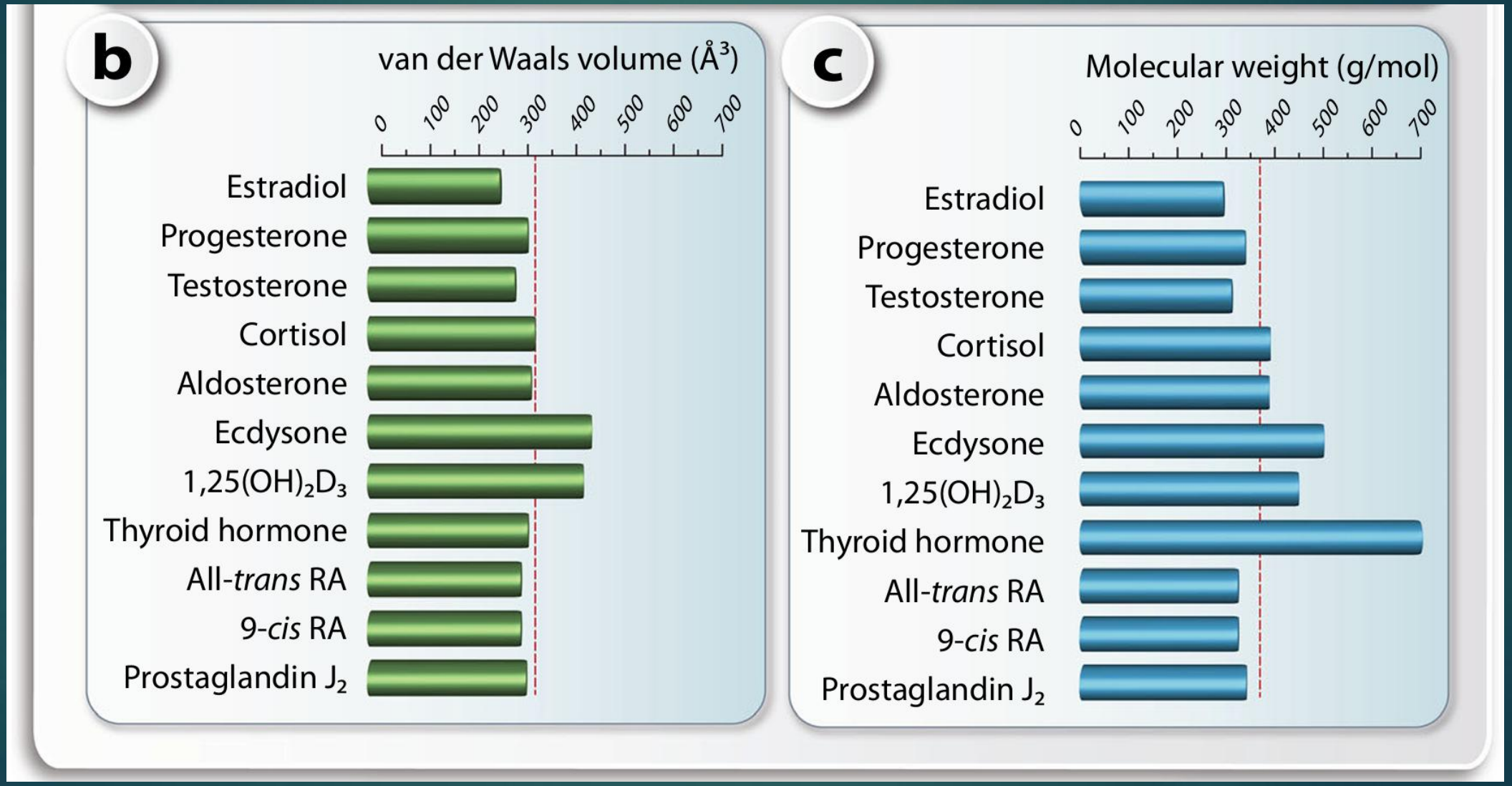
Molecular Interactions of Nuclear Receptor

Pocket Volume Differences

- Endocrine Receptors:
  - Moderate volume (300–700 Å^3)
  - ligand fills 60–80% of the pocket (high specificity/affinity).
- Adopted Orphans:
  - Large volume (up to 1400 Å^3)
  - Ligand fills only 25–50%, allowing binding of diverse promiscuous ligands.



Molecular Interactions of Nuclear Receptor



DNA Binding Specificity: The "1-to-5 Rule"

- Dimerization
  - Most nuclear receptors bind DNA as dimers.
  - Homodimers: Steroid receptors (GR, AR, MR, PGR) bind inverted repeats of hexameric motifs.
  - Heterodimers: 14 receptors (including PPAR, LXR, VDR, THR, RAR) obligatorily partner with RXR (Retinoid X Receptor).
- The 1-to-5 Rule
  - RXR heterodimers bind **Direct Repeats (DR)** of the motif AGGTCA.
  - Specificity is determined by the number of spacer nucleotides:
    - DR1: PPAR-RXR (Peroxisome Proliferator-Activated Receptor)
    - DR2: (Less common)
    - DR3: VDR-RXR (Vitamin D Receptor)
    - DR4: THR-RXR (Thyroid Hormone Receptor)
    - DR5: RAR-RXR (Retinoic Acid Receptor)

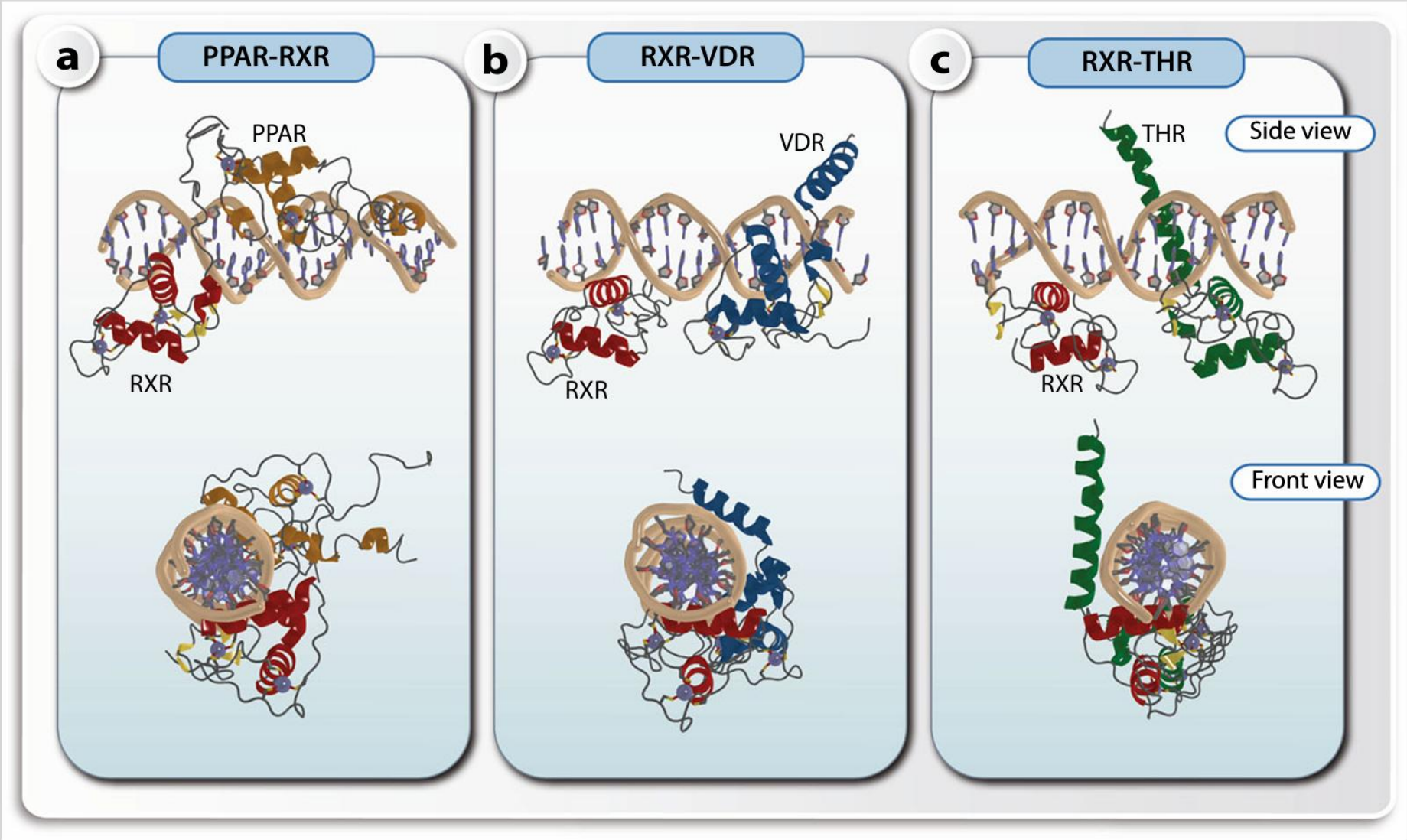
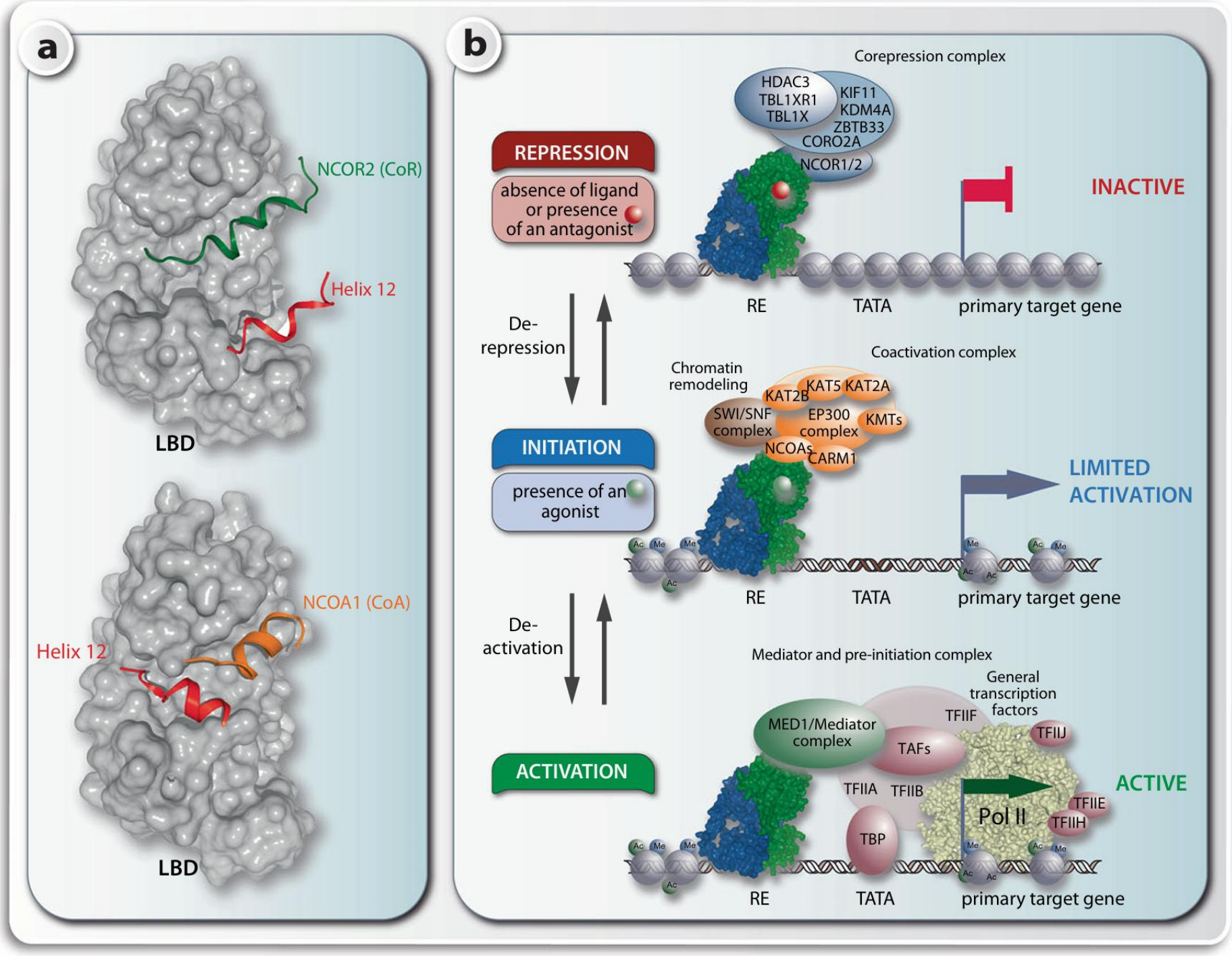


Fig. 5.4 Binding of RXR heterodimers to DR-type REs. Heterodimeric DBD complexes of RXR with PPAR on a DR1-type RE **a**, with VDR on a DR3-type RE **b** and with THR on a DR4-type RE **c** are displayed in two different orientations

Mechanism of Transcriptional Activation

The 3-Step Transactivation Process

- Repression (Absence of Ligand):
  - DNA-bound nuclear receptors recruit Corepressors (e.g., NCOR1, NCOR2).
  - These complexes recruit HDACs (Histone Deacetylases), leading to local chromatin condensation and gene silencing.
- Activation (Ligand Binding):
  - Ligand binding triggers a conformational change (Helix 12 flip), displacing corepressors.
  - Coactivators (e.g., NCOA family) are recruited, which bring in HATs (Histone Acetyltransferases) to open the chromatin (de-repression).
- Initiation:
  - Coactivators are replaced by the Mediator (MED) complex.
  - This connects the receptor to the basal transcriptional machinery (Pol II), initiating transcription.

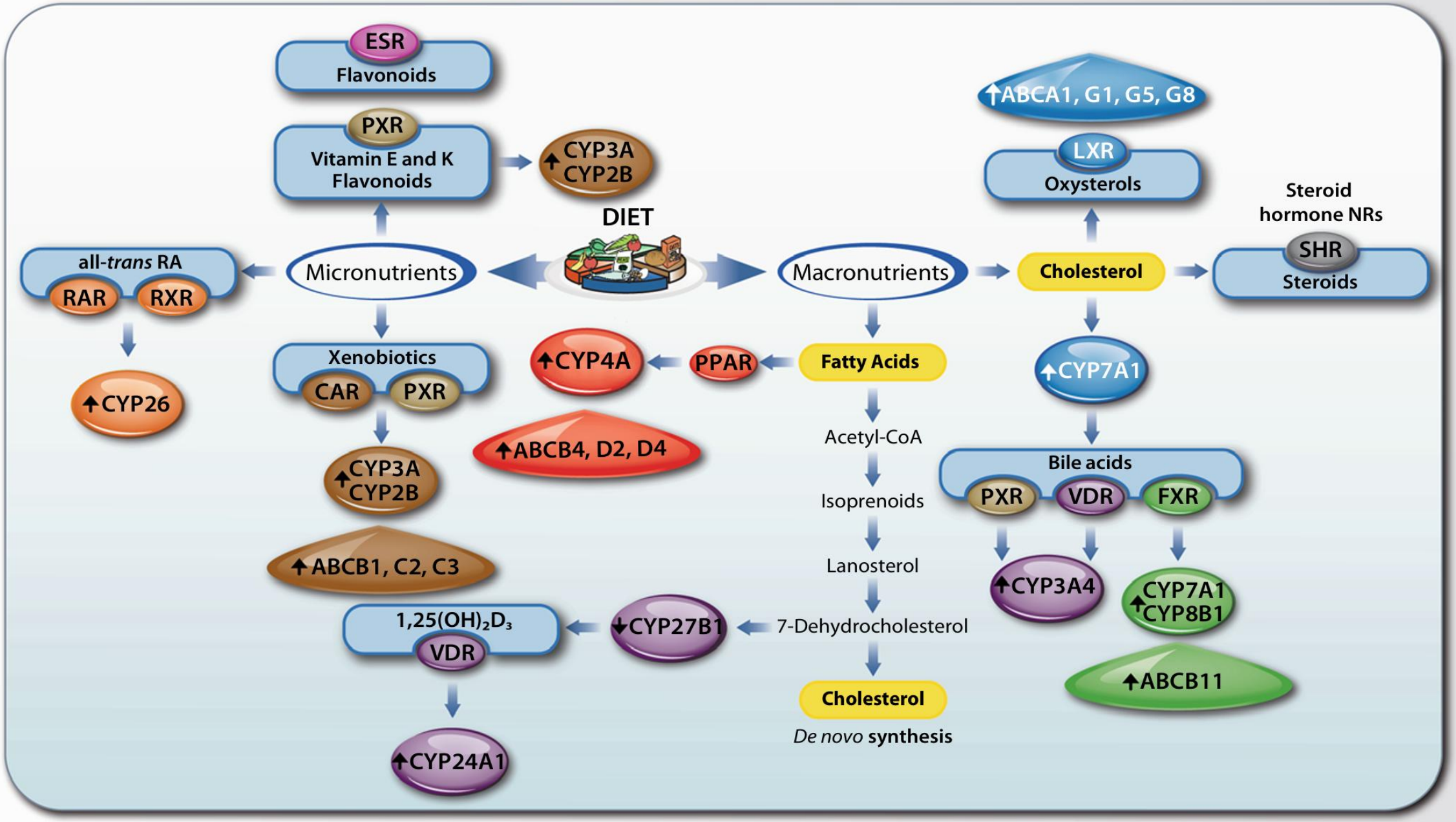


Interaction of nuclear receptors with cofactors.
A solvent excluded surface (Connolly surface) representation of a nuclear receptor LBD in the absence (top) and presence (bottom) of a ligand

Nuclear Receptors as Nutrient Sensors

Metabolic Integration

- **Adopted orphan receptors** act as sensors for micro- and macronutrients, regulating lipid, glucose, and energy metabolism.
- **Triangle Regulatory Circuits**: A metabolite activates a receptor → receptor regulates the enzyme/transporter → enzyme controls metabolite levels.


ESR
Flavonoids
PXR
Vitamin E and K
Flavonoids
↑CYP3A
CYP2B
DIET
all-trans RA
RAR
RXR
Micronutrients
Macronutrients
↑ABCA1, G1, G5, G8
LXR
Oxysterols
Steroid
hormone NRs
SHR
Steroids
Cholesterol
↑CYP26
Xenobiotics
CAR
PXR
↑CYP4A
PPAR
Fatty Acids
↑CYP7A1
↑ABCB4, D2, D4
Acetyl-CoA
Bile acids
PXR
VDR
FXR
↑CYP3A
CYP2B
Isoprenoids
↑ABCB1, C2, C3
Lanosterol
↑CYP3A4
CYP7A1
↑CYP8B1
1,25(OH)₂D₃
VDR
↓CYP27B1
7-Dehydrocholesterol
↑ABCB11
Cholesterol
De novo synthesis
↑CYP24A1

Nuclear Receptors as Nutrient Sensors

Key Metabolic Receptors:

- PPARs (Fatty Acid Sensors):
  - PPAR alpha: Liver/heart. Regulates fatty acid oxidation (fasting response) and ketogenesis.
  - PPAR gamma: Adipose tissue. Master regulator of adipogenesis and insulin sensitivity.
- LXRs (Oxysterol Sensors):
  - Promote reverse cholesterol transport (via ABCA1, ABCG1) to eliminate excess cholesterol.
  - Stimulate lipogenesis (SREBF1, FASN) to buffer free cholesterol.
- FXR (Bile Acid Sensor): Protects liver from bile acid toxicity by inhibiting their synthesis (CYP7A1) and promoting excretion.

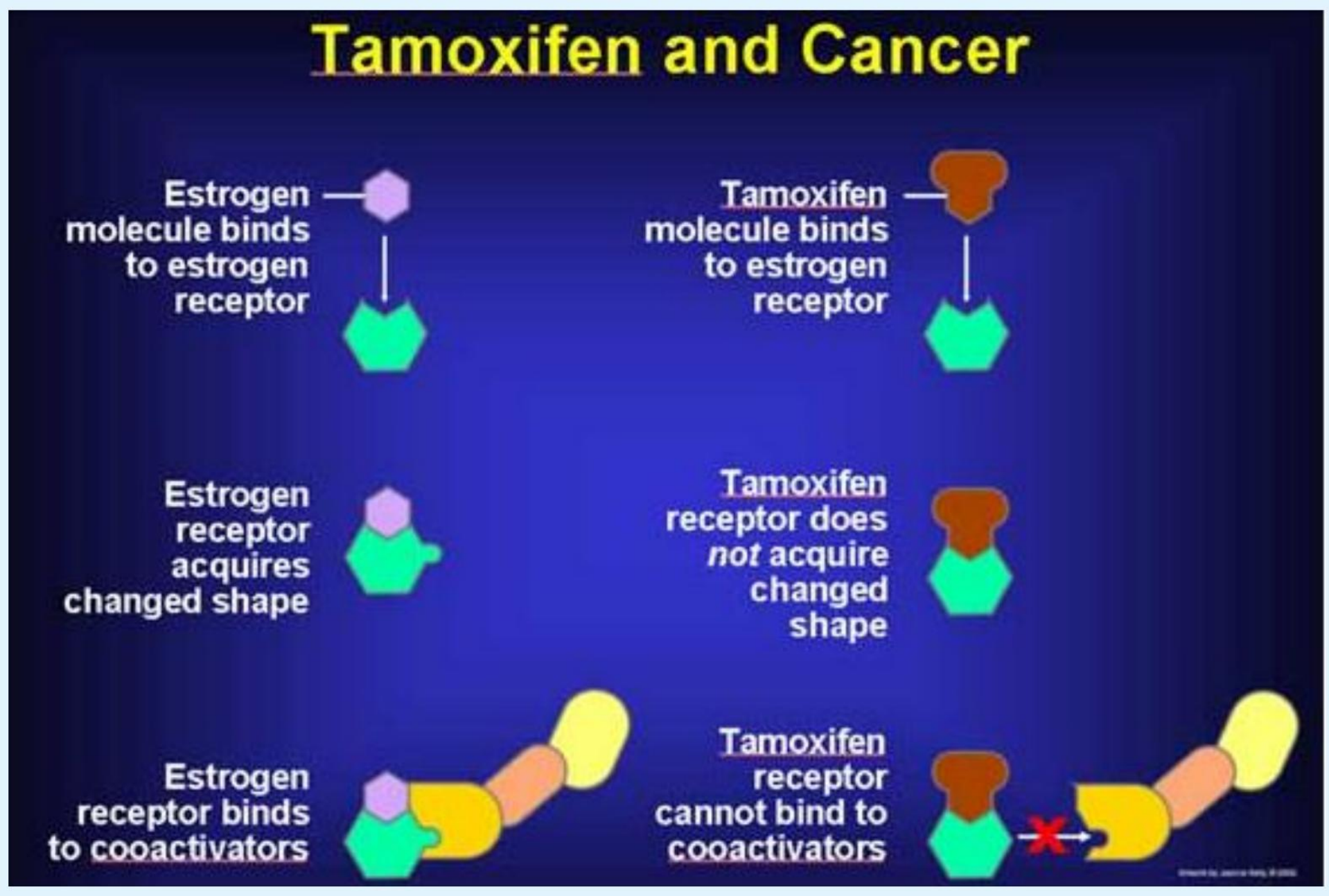
Clinical Relevance and Therapeutics

Pharmacology

- Nuclear receptors are major drug targets due to their lipophilic ligand pockets.
- Thiazolidinediones (e.g., Rosiglitazone): Activate PPAR-gamma to treat Type 2 Diabetes (improve insulin sensitivity).
- Fibrates: Activate PPAR-alpha to treat hyperlipidemia.
- Tamoxifen: ER antagonist used in breast cancer therapy.

Inflammation and Immunity

- Many NRs (GR, VDR, PPAR, LXR) exert anti-inflammatory effects.
- Transrepression: They inhibit inflammatory transcription factors like NF-kappaB and AP1 through protein-protein interactions, reducing cytokine production.
- Vitamin D (VDR): Promotes antimicrobial peptide expression (CAMP) in macrophages to fight infection (e.g., tuberculosis).



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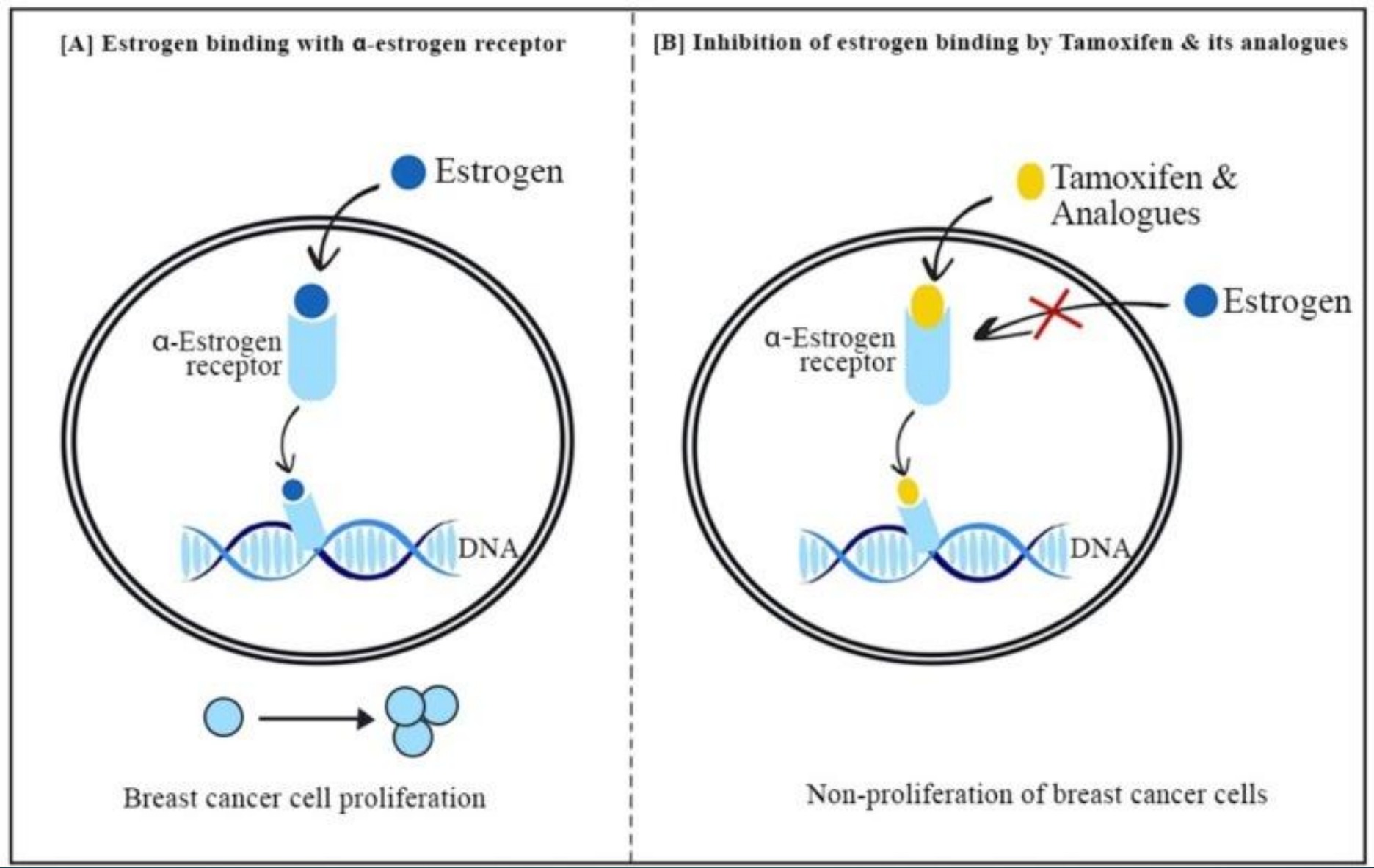

10.1016/j.molliq.2024.126209

Summary

- Nuclear receptors serve since some 30 years as paradigms for many functional and structural aspects of transcription factors, such as the interaction with coactivator and corepressor proteins.
- Many of the 48 human members of the largest family of transcription factors in metazoans have the unique property to be specifically activated by small lipophilic ligands in the size of cholesterol (approximately 400 Da).
- Some of these ligands are known as endocrine hormones, such as estradiol and testosterone, while others are metabolites of dietary compounds, such as fatty acids and cholesterol.
- Both types of lipids are of large physiological impact in health and disease, such as metabolic control and the modulation of immunity.
- This made nuclear receptors especially attractive for both basic and applied research.

Xin chân thành cảm ơn!

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